Supplemental Material for: Combined topological and Landau order from strong correlations in Chern bands

Stefanos Kourtis and Maria Daghofer
Institute for Theoretical Solid State Physics, IFW Dresden, 01171 Dresden, Germany

(Dated: October 2, 2014)

We give here data on the impact of longer-range V_2 and V_3 on the topological pinball liquid (TPL). We also shortly mention thermodynamic stability and larger lattices.

PACS numbers: 71.27.+a, 03.65.Vf, 71.45.Lr, 73.43.-f

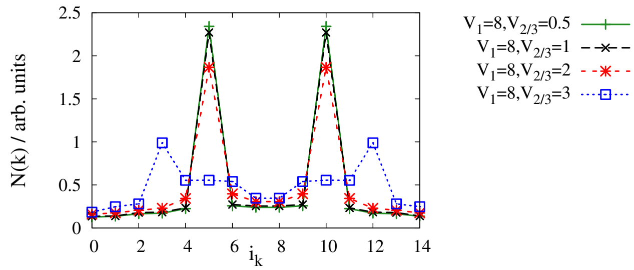

FIG. 1. (Color online) Static charge-structure factor $N(\mathbf{k})$ illustrating how longer-range coulomb repulsion $V_2 = V_3$ destabilizes the CDW pattern shown in Fig. 1(b) of the main text. Momenta with numbers 5 and 10 correspond to $\pm\mathbf{K}$, the ordering momentum of the CDW, for $V_2 = V_3 = 3t$, $N(\mathbf{k})$ is no longer peaked here. $V_1 = 8t$ and $t' = 0.2t$ in all cases, filling is 12 electrons on 30 sites/15 unit cells.

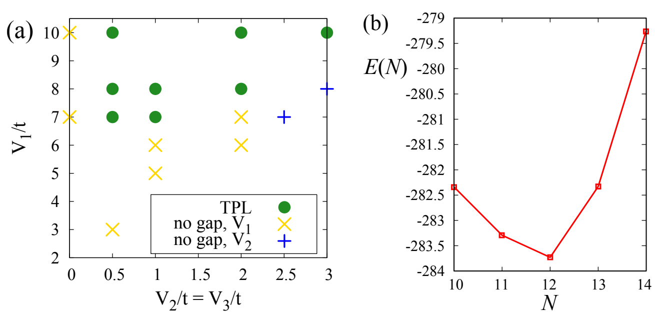

FIG. 2. (Color online) (a) Phase diagram for $\bar{n} = 12/30$ depending on V_1 and $V_2 = V_3$ for $t' = 0.2t$. Filled circles denote TPL states. $\times$ denote non-TPL states, which do not show a gap between the lowest 15 states and the rest of the spectrum for all fluxes $\varphi = (\varphi_{\mathbf{a}_2}, \varphi_{\mathbf{a}_3})$, but where $N(\mathbf{k})$ is still peaked at $\pm\mathbf{K}$. $+$ denote non-TPL states where these peaks in $N(\mathbf{k})$ have been lost, i.e., where V_2 destabilizes the charge pattern, see Fig. 1. (b) Total energy as a function of particle number on the 30-site cluster for $V_1/t = 10, V_2/t = V_3/t = 2$.

Figure 1 shows that $V_2 = V_3 \gtrsim 3t$ destroys the CDW depicted in Fig, 1(b) of the main text for $V_1 = 8t$. Generally, the critical value is $V_2 \approx V_1/3$. As can be seen in the phase diagram for $t' = 0.2t$ in Fig. 2(a), finite V_2 and V_3 is helpful, and may even be necessary, in inducing the TPL. When longer-range interactions are strong enough to weaken the CDW, however, first the TPL disappears and $N(\mathbf{k})$ finally is no longer peaked at $\pm\mathbf{K}$.

Phase separation into a charge ordered and an FCI domain within the lattice would be hard to reconcile with the observables discussed in the main text. Moreover, we checked that the total energy depending on filling is convex for fillings between 10 and 14 electrons on the 30-site cluster, see Fig. 2(b), also arguing for a thermodynamically stable phase.

For a filling of 18 electrons on 30 sites, we likewise found (i) charge order, (ii) a 15-fold near ground-state degeneracy, and (ii) $\sigma_H = -0.4$ (with $V_1/t = 20$, $V_2/4 = V_3/4 = 4$, $t'/t = 0.2$). This agrees with expectations based on a particle-hole transformed situation of the 12-electron case discussed in the main text: 10 holes form the CDW while the remaining 2 move between them. This supports our picture of the topological pinball liquid, as the "normal" pinball liquid is stable for $1/3 < \bar{n} < 2/3$ and extends the stability range of the TPL.

While a finite-size scaling involving several cluster sizes at the same filling is not possible in our case, we were able to address somewhat larger clusters for the limit of strong nearest-neighbor Coulomb repulsion. In this limit, the Hilbert space can be reduced by discarding high-energy states with too many electrons occupying NN sites, i.e., only states with an 'almost perfect' CDW have to be kept. We can then treat a cluster with 6×6 sites (18 unit cells) and a filling of 14 electrons. We allowed 0, 1, or 2 faults in the CDW, by finding consistent results, we can conclude that the approximated Hilbert space still captures all relevant states. In agreement with the expectations for a topological pinball liquid, we find a nine-fold degenerate ground state, 3-fold for the CDW and 3-fold for the FCI. The latter can be understood in terms of two electrons that fill one third of the lowest subband resulting from the effective lattice. Accordingly, we find $\sigma_H = 1/3$ in each of the nine states and inserting fluxes reveals that three groups of three states each show spectral flow, with $6\pi = 3 \times 2\pi$ bringing the system back to the original point. Parameter sets treated include $V_1/t = 100, V_2/t = V_3/t = 1$, $V_1/t = 100, V_2/t = V_3/t = 2$, $V_1/t = 10, V_2/t = V_3/t = 2$; $t'/t = 0.2$ in all cases.

The above results also further ascertain the robustness of the CDW part of the TPL states as the system size is increased. Taking this robustness in the strong-V_1 limit into account, we can restrict the Hilbert space to states that contain perfect charge order, as any faults in the charge pattern will be severely penalized energetically. As detailed in the main text, the charge order is associated with a 3-fold

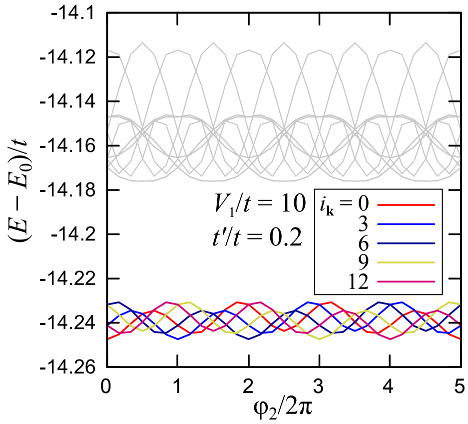

FIG. 3. Spectral flow of the eigenvalues of the effective honeycomb lattice [denoted by thick black lines in Fig. 1(b) of the main text] with 6 particles in a 3×5-cell cluster ($\nu = 2/5$) with $V_1/t = 10$ and $t'/t = 0.2$, which models the residual lattice after assuming perfect charge order.

ground-state degeneracy, which arises from the translations of the charge pattern. Moreover, when the strong-V_1 limit suppresses deviations from perfect charge order, it also removes tunneling between the three configurations of the CDW and the Hilbert space consequently decomposes into three blocks. Treating each of these blocks corresponds to keeping only the states coming from one of the copies of the CDW pattern. The "pins" of the pinball can thus be kept fixed and the interacting "residual" particles can be treated on the effective honeycomb-lattice model of Fig. 1(b) of the main text.

For this model, we have studied the $\nu = 2/5$ state for up to 6 particles. We have verified that the ground state is 5-fold degenerate (since we have removed by hand the 3-fold degeneracy of the CDW), the levels that comprise it exhibit spectral flow (see Fig. 3), and the corresponding states have a very precicely quantized $\sigma_{\mathrm{H}} = 2/5$ in units of e^2/h. Therefore, as long as the CDW component of a TPL state remains robust when the system size is increased (as was seen above), it is reasonable to expect that TPL states survive for larger systems as well. Phase separation into a charge ordered and an FCI domain within the lattice would be hard to reconcile with the observables discussed above. Additionally, the total energy depending on filling is convex for fillings between 10 and 14 electrons on the 30-site cluster, as shown in Fig. 2(b), also arguing for a thermodynamically stable phase. This suggests that TPL states will be, in principle, stable in the thermodynamic limit, at least in a perfect system at zero temperature, since there is no intrinsic feature of these states that prevents thermodynamic stability.